

3rd Homework sheet (due Oct 10)

k always denotes an algebraically closed field of characteristic 0.

Problem 1 (Constructible sets, 5P)

Let X be a variety and let $W \subseteq X$ be a constructible subset. Show that there exists a subset $U \subseteq W$ which is open and dense in the closure $\overline{W}$.

Problem 2 (Homomorphisms to $\mathbb{G}_a$, 2P + 2P)

Let $(G, m) = \mathrm{Sp}(A, m^*)$ be a LAG over k .

(a) Show that the homomorphisms $G \rightarrow \mathbb{G}_a$ are in bijection with the elements $a \in A$ such that $m^*(a) = a \otimes 1 + 1 \otimes a$.

(b) Show that there are no non-zero homomorphisms $\mathrm{GL}_n \rightarrow \mathbb{G}_a$.

Problem 3 (Constructing PGL_n , 2P + 3P + 3P)

Consider the special linear group $\mathrm{SL}_n = \mathrm{Sp}(A)$,

$$A = k[x_{11}, \dots, x_{nn}] / (\det(x_{ij}) - 1).$$

(a) Check that polynomial degree induces a $\mathbb{Z}/n\mathbb{Z}$ -grading $A = \bigoplus_{r \in \mathbb{Z}/n\mathbb{Z}} A_r$ on A .

This means that each A_r is a k -subvector space and $A_r \cdot A_s \subseteq A_{r+s}$.

(b) Prove that the ring of degree 0 elements $A_0 \subset A$ is a finitely generated k -algebra.

(c) Show that A_0 is stable under comultiplication, i.e. that $m^*(A_0) \subseteq A_0 \otimes A_0$. Deduce that $\mathrm{Sp}(A_0)$ is naturally a LAG with group structure coming by restriction from m^* .

At this point, you have defined the *projective linear group* $\mathrm{PGL}_n = \mathrm{Sp}(A_0)$ as a LAG! It can be written in the two ways

$$\mathrm{GL}_n / \mathbb{G}_m \quad \text{or} \quad \mathrm{SL}_n / \mu_n.$$

The exercise makes the second description explicit.

Problem 4 (Translation invariant functions on $\mathbb{G}_m$, 2P + 3P)

Consider the multiplicative group $\mathbb{G}_m = \mathrm{Sp}(A)$, $A = k[t, t^{-1}]$.

(a) For each $1 \leq d$, let $F_d \subseteq A$ be the $(d+1)$ -dimensional subspace of polynomials in t of degree $\leq d$. Show that F_d is translation invariant.

(b) Choosing a basis $g_0, \dots, g_d$ of F_d as in the proof of the embedding theorem during lecture, describe the corresponding representation

$$\mathbb{G}_m \longrightarrow \mathrm{GL}_{d+1}.$$